

IEDA 4000H: Optimal Rebalancing Research

Project Report

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Abstract

This research project investigates the development of optimal portfolio rebalancing strategies, focusing on the critical trade-off between maintaining target allocations and minimizing transaction costs. Through a combination of theoretical derivation, comprehensive simulations, and sensitivity analysis, we evaluate the performance of periodic and threshold-based rebalancing mechanisms under varying market conditions and cost structures. Our findings demonstrate that, under typical institutional cost assumptions, a theoretically optimized periodic schedule (e.g., monthly) provides superior economic efficiency and net returns compared to a dynamic threshold strategy, which incurs excessive trading costs despite tighter allocation control. The analysis further reveals the regime-dependent nature of rebalancing efficacy and highlights the importance of accurately modeling cost structures (proportional vs. fixed). We conclude that while threshold rebalancing offers precise allocation management, its practical value is often eroded by costs, making periodic rebalancing a more robust and implementable baseline for most institutional investors.

1 Introduction

Portfolio rebalancing, as defined by Karabatı [1], involves the regular adjustment of asset weights within a portfolio to maintain its intended performance and risk profile. This process is crucial because market fluctuations inevitably alter the original allocation of assets. By periodically adjusting these proportions, investors ensure their portfolio continues to align with their pre-determined risk tolerance and investment objectives [1]. This is particularly important as the effectiveness of a rebalancing strategy is inherently tied to an investor’s risk appetite, creating a crucial trade-off with the associated transaction costs [1].

Intuitively, a more risk-averse investor might rebalance more frequently or aggressively to quickly bring their portfolio back to its target allocation, thereby reducing their exposure to market volatility. Conversely, a more risk-tolerant investor might rebalance less often, accepting larger deviations in asset weights in pursuit of potentially higher returns. However, this intuitive understanding highlights a fundamental challenge: while rebalancing aims to mitigate risk and maintain alignment with investment goals, frequent rebalancing incurs additional costs, such as commissions and bid-ask spreads. This delicate balance between risk management and cost minimization is a central dilemma in portfolio management. Beyond its role in risk control and alignment, rebalancing serves as a vital tool for mitigating the impact of behavioral biases [2]. By enforcing a disciplined approach, it counters common emotional pitfalls like performance chasing during bull markets and panic selling during downturns, thereby promoting more rational investment decisions.

Given these demonstrated benefits of rebalancing and the inherent trade-offs involved, this paper investigates the development of optimal rebalancing strategies, specifically considering the impact of incurred transaction costs. We delineate the main general categories of rebalancing strategies into three primary approaches: periodic rebalancing (PR), which involves rebalancing at fixed intervals such as quarterly or weekly; threshold (or tolerance-band) rebalancing (TR), where adjustments are triggered when asset weights deviate beyond pre-defined thresholds; and a combination of both PR and TR [3], [4]. A compelling argument for a more dynamic approach is presented by Daryanani [5], who advocates for “opportunistic rebalancing.” This strategy emphasizes continuous monitoring and adjusts the portfolio only when necessary, offering significant advantages over traditional, rigid annual or quarterly schedules. Daryanani [5] further posits that, beyond its risk-control ca-

pabilities, opportunistic rebalancing actively enhances returns by strategically capitalizing on sporadic buy-low/sell-high opportunities within the market [1].

2 Literature Review

A review of the existing literature was conducted to establish a foundation for analyzing optimal portfolio rebalancing. Three pivotal studies were examined in detail due to their significant contributions to this field. The following summary outlines their key findings and relevance to our research.

2.1 Literature Review on the given article

1. Introduction

The optimal rebalancing frequency represents a fundamental trade-off in portfolio management. Frequent rebalancing allows portfolios to maintain the desired target weights of assets, potentially reducing tracking error, but incurs higher transaction costs that can erode performance. The availability of high-frequency financial data has introduced a new layer of complexity to the classic covariance estimation problem. In theory, realized volatility and covariance estimators built from intraday returns offer significantly greater precision. In practice, however, their effectiveness hinges on how these improved estimates are incorporated into the portfolio rebalancing process. Liu (2009) addresses this problem by how the economic benefits of high-frequency data vary with rebalancing frequency. This study demonstrates that the value of high-frequency data is not absolute but conditional on the rebalancing strategy employed by the investor.

2. Methodology

Liu [6] provided a methodological framework for disentangling the effects of portfolio rebalancing and data frequency on tracking error. The study simulates a professional portfolio manager’s objective to replicate the S&P 500 index by constructing a minimum tracking-error-volatility portfolio using the 30 Dow Jones Industrial Average (DJIA) stocks. The dataset consists of high-frequency, 5-minute returns from the TAQ and CME databases, as well as daily returns, for the period 1993–2000, is well-suited for the analysis, as the

highly liquid DJIA constituents, which represent approximately 20% of the total U.S. stock market value, exhibited a median inter-trade time of under one minute, ensuring the reliability of the high-frequency estimates.

The study constructs and compares a wide range of covariance matrix estimators. They are categorized into two major types, Low-Frequency Estimators (Daily Data) and High-Frequency Estimators (Intraday Data). Low-Frequency estimators include the sample covariance matrix, versions adjusted for lead-lag effects, an optimal shrinkage estimator, and an optimal rolling sample estimator. In contrast, high-frequency estimators encompass another broad range of techniques, including basic estimators, versions incorporating overnight returns, adjustments for intraday volatility patterns, corrections for first-order autocorrelation, and Optimal Rolling Sample Covariance Matrix Estimator. The core of Liu's methodology is the systematic variation of two parameters, rebalancing frequency and estimation variable. The portfolio is rebalanced either monthly or daily to examine the impact of rebalancing timing, while the length of historical data used to estimate the covariance matrix varies from 1 month to 12 months. This tests the robustness of different estimators when the amount of available data is limited.

The out-of-sample performance is evaluated by the annualized standard deviation of the tracking error. Additionally, the study assesses the practical feasibility of the strategies by analyzing portfolio turnover and the impact of proportional transaction costs.

3. Major Findings

The study shows the effectiveness of rebalancing strategies depends on the data choice. When the portfolio is rebalanced monthly using a 12-month estimation window, the study finds no significant benefit from using high-frequency data. With infrequent rebalancing, the high-frequency information's advantage in precision is "averaged out" over the longer holding period and cannot be actively exploited. In contrast, when the portfolio is rebalanced daily, substantial improvements emerge from using high-frequency data. Notably, the rolling-sample realized covariance estimator, which incorporates both overnight returns and lead-lag adjustments, delivers a tracking-error volatility of just 4.06%, significantly and statistically lower than the 4.44% achieved by the best daily-data estimator. Daily rebalancing allows the manager to continuously update the portfolio to reflect the more accurate,

daily covariance estimates provided by high-frequency data.

The benefits of high-frequency data are also pronounced when the estimation horizon is short. If a manager has access to less than 6 months of data, the performance of daily-data-based estimators deteriorates rapidly as the window shortens, while high-frequency data provides a dense information set that can compensate for a short historical time series, which is particularly valuable for new assets or during structural breaks in the market. For instance, with only one month of daily data, the tracking error volatility becomes extremely high (218.75% annually). However, estimators based on just one month of high-frequency data remain stable and effective (around 4.6% volatility).

Another finding in the literature concerns the significant impact of covariance matrix estimation on portfolio turnover and effect of transaction cost. Research demonstrates that estimators derived from high-frequency data yield portfolios with substantially lower turnover rates compared to those based on daily returns, a benefit that persists across both daily and monthly rebalancing frequencies. This stands in contrast to rolling-window estimators, which, by overweighting recent information, provoke more aggressive weight updates and consequently higher turnover. The practical advantage of high-frequency methods becomes most apparent after accounting for transaction costs. While such costs uniformly diminish mean returns, their effect on tracking error volatility is negligible. Crucially, for daily rebalancing, strategies employing high-frequency data, particularly those incorporating overnight returns and lead-lag effects, maintain their viability. They exhibit a comparable tracking error but a smaller decline in mean return than the daily-returns benchmark, ensuring that the performance gains are not eroded by transaction costs. The consensus, therefore, is that high-frequency data renders minimum-variance strategies more practical and cost-effective by curbing turnover without sacrificing performance benefits, an advantage that remains robust even under short-sale constraints.

4. Conclusion

The study by Liu [6], demonstrates that the question is not simply whether high-frequency data is better, but under what operational conditions it provides economic value. Rebalancing frequency is a decisive factor in this case. The benefits of high-frequency data for portfolio optimization are primarily realized in strategies involving frequent rebalancing

(e.g., daily) or when the estimation horizon is short (less than 6 months).

2.2 Literature Review of Additional Source 1

1. Introduction

Bağcı and Soylu [7] explore optimal rebalancing strategies within the context of high-frequency trading (HFT), focusing on a specific application: pair trading with a threshold-based rebalancing algorithm (HFRA). Their research, titled "Optimal portfolio selection with volatility information for a high frequency rebalancing algorithm," investigates the efficacy of this strategy in various market scenarios, particularly concerning its long-term profitability and sustainability. The core inquiry revolves around identifying the optimal combination of threshold levels, portfolio size, and diversification that yields the best outcomes for this HFRA strategy when applied to a set of 40 crypto assets.

2. Methodology

The study's methodology employs the Johansen cointegration test and a Vector Error Correction Model (VECM) to establish and ensure long-term, statistically stable pair trading relationships between the crypto assets within the portfolio. This crucial step ensures that the foundation of the pair trading strategy—the mean-reverting property of the asset spread—is robust. Upon establishing these relationships, a threshold rebalancing logic, explicitly detailed in the study's pseudocode, is implemented to trigger trades when the price spread between the co-integrated assets deviates beyond a predefined threshold. The objective of this simulation-based study, utilizing closing price data per minute in 2021 from 40 crypto assets with the highest market capitalization on Binance, is to maximize potential returns and minimize potential losses across different market scenarios. Key variables analyzed include the rebalancing threshold TTT, portfolio size, and the number of assets in the portfolio, with performance evaluated based on portfolio returns/losses by the end of the period.

3. Major Findings

The findings from Bağcı and Soylu [7] offer significant insights into the performance

of HFRA strategies. The HFRA consistently demonstrated superior performance compared to both periodic and normal threshold rebalancing strategies across all analyzed market trends. More specifically, in up-trend markets, the strategy's strongest performance was observed in high-volatility scenarios, while low-volatility conditions limited trading opportunities and resulted in reduced gains. Conversely, in down-trend markets, high volatility allowed for some loss mitigation through timely rebalancing, showcasing the strategy's protective capabilities. However, consistent losses were observed during steady declines in low-volatility environments. Furthermore, in no-trend markets, high volatility enabled modest profitability, a performance that generally improved with increased portfolio size. In contrast, low-volatility conditions in no-trend markets yielded only minimal returns, regardless of diversification levels.

Volatility emerged as a critical factor, significantly influencing the strategy's ability to yield returns and prevent losses. The high-frequency nature of the strategy allowed it to capitalize on price divergences that occur more frequently and with greater magnitude during volatile periods, serving as a primary driver for the observed outperformance.

The impact of portfolio size was contingent on market volatility. During up-trends with high volatility, a larger portfolio size significantly boosted returns, improving performance from 151.5% to 160.46%. In down-trends with high volatility, larger portfolios were found to reduce losses, indicating improved loss prevention from -25.20% to -23.04%, likely attributable to better diversification within the larger set of assets. However, under no-trend and low-volatility conditions, portfolio size had a minimal impact due to limited price divergences and, consequently, fewer trading opportunities.

Similar to portfolio size, the number of assets played a crucial role primarily in high-volatility conditions, where it enhanced overall performance by providing more opportunities for pair trading and diversification. In low-volatility environments, however, where price divergences are limited, the number of assets had a minimal effect on the strategy's outcome.

4. Conclusion

In summary, Bağcı and Soylu [7] highlight that for a high-frequency threshold-based

pair trading strategy in the crypto market, volatility is a paramount driver of profitability and loss prevention. The strategy thrives in volatile conditions, where it leverages price divergences. Optimal portfolio size and diversification, particularly the number of assets, are also dependent on market volatility, with their benefits becoming more pronounced in high-volatility settings where more trading opportunities arise. This study underscores the potential of adaptive, high-frequency rebalancing strategies, especially when integrated with volatility information, to outperform traditional methods in dynamic asset classes like cryptocurrencies.

2.3 Literature Review of Additional Source 2

1. Introduction

While high-frequency data and dynamic rebalancing have been explored in related literatures, this review focuses on portfolio rebalancing with low-frequency data and strategic timing optimization. Karabatı [8] proposes a robust optimization model that operates on monthly data and determines a fixed number of optimally timed rebalancing points within an annual horizon. Unlike high-frequency or strictly calendar-based methods, this framework introduces robustness by evaluating rebalancing schedules across multiple historical market regimes, seeking to enhance returns without increasing rebalancing frequency. This review examines the methodological foundations and contributions of this low-frequency, optimization-driven approach to portfolio rebalancing, with particular attention to its data requirements, timing mechanism, and practical implementation.

2. Methodology

The methodological framework proposed by Karabatı [8] is built upon three interconnected modeling components. The foundation is a network flow model that structures the rebalancing timing problem. This model represents an investment horizon as a directed acyclic graph, where nodes correspond to potential rebalancing months and arcs represent the logarithmic return achieved by holding a portfolio formed at the start node for a specific number of months. The core problem is formulated as a longest-path optimization with a side constraint that fixes the total number of rebalancing events. This allows the problem to be transformed into a shortest path problem, which can be solved more easily. The model's

parameters are derived from an underlying portfolio formation strategy, specifically a standard Mean-Variance optimizer, which is executed using low-frequency monthly return data from a predefined set of assets. This first model is a self-contained tool for backtesting an optimal rebalancing strategy for a single historical period.

To prevent overfitting to a specific historical regime, the paper introduces a second, robust optimization model. This model generalizes the analysis by considering multiple overlapping historical periods, illustrated as "time blocks." For each block, the model first calculates the block-specific optimal rebalancing strategy. The robust model then seeks a single unified rebalancing schedule that minimizes the maximum performance deviation from each block's ideal outcome. This min-max objective function incorporates uncertainty about future market conditions by ensuring the chosen strategy performs adequately across diverse past environments. Together, the two models form a flexible, data-driven system for determining rebalancing time points that are both historically optimal and resilient to market variability.

3. Major Findings

The computational evaluation of Karabati's robust optimization framework reveals three principal findings that underscore the value of its low-frequency, timing-oriented methodology.

The paper shows significant and robust performance improvement when compared to traditional calendar rebalancing. Using monthly data and a fixed rebalancing frequency of four times per year, the robust timing strategy generated a statistically significant improvement in average annual returns compared to standard quarterly rebalancing. The out-of-sample tests over the 1971–2020 period showed consistent added value, with annual returns increasing by approximately 1.6% to 2.7% above the quarterly benchmark. Moreover, due to the effect of compounding, this annual improvement translated into a terminal wealth ratio of 2 to 3 times that of the quarterly strategy over the 50-year horizon. Since both strategies rebalance the same number of times annually, this increase in return can be accounted to smarter rebalancing timing rather than different trading costs or frequency.

Another finding of the literature is that the framework's performance is sensitive to the

construction of the historical “analysis window.” An effective window length of approximately 6–7 years, structured as multiple overlapping time blocks, yielded the most reliable results. This finding highlights that robustness is not achieved by simply using more data, but by strategically evaluating timing decisions across varied historical regimes. The multi-block robust optimization successfully mitigates the risk of overfitting to a single market period, confirming that the methodology’s core innovation—optimizing against worst-case historical regret—effectively enhances out-of-sample resilience.

4. Conclusion

This robust optimization framework demonstrates that a systematic, data-driven approach to rebalancing timing can significantly enhance portfolio performance, even when rebalancing frequency remains fixed. By modeling the problem as a network flow with a multi-block robust criterion, the method identifies historically resilient timing points using only low-frequency data. The key insight for designing an optimal rebalancing strategy is clear: strategic timing, informed by robust multi-regime backtesting, can be a critical source of added value, offering a disciplined alternative to arbitrary calendar-based rules. The framework provides a practical blueprint for developing a rebalancing strategy that prioritizes timing optimization and robustness over mere frequency.

2.4 Literature Comparison and Analysis

The three literatures analyzed previously presents diverse methodologies, each tailored to specific market conditions, data environments, and investment horizons. The studies examined offer distinct yet complementary approaches, ranging from high-frequency tactical adjustments to low-frequency strategic optimization. This section compares these frameworks across their core strategies, suitable applications, and practical implications to delineate their respective contributions and limitations.

1. Study overview

The three studies collectively demonstrate a spectrum of solutions, ranging from tactical, high-frequency algorithmic design to strategic, low-frequency optimization. Their

primary goals and innovations are fundamentally shaped by their targeted investment horizon, proving that no single approach is universally applicable.

Comparative Dimension	Liu (2009) (Provided paper)	Bağcı & Soylu (2024) (Review 1)	Karabatı (2025) (Review 2)
Primary Goal	Evaluate when HF data improves covariance estimation for minimum-volatility portfolios.	Propose a high-frequency algorithm for capturing profit.	Determine the optimal timing of a fixed number of rebalancing events over a long horizon.
Investment Horizon	Daily Rebalancing	Intraday	Long-Term (Annual Horizon)
Focus	Pragmatic assessment of HF data utility vs. estimation window trade-offs.	Dynamic threshold-based triggers within a pair trading framework.	Robust optimization using historical regime analysis to select calendar dates.

2. Methodology

The choice of data frequency and core logic is a strategic decision that must align with the method’s ultimate objective. Whether the driver is capturing intraday signals, improving risk estimation, or achieving long-term outperformance, each methodology employs a distinct optimization framework tailored to its specific purpose.

Comparative Dimension	Liu (2009) (Provided paper)	Bāğcı & Soylu (2024) (Review 1)	Karabatı (2025) (Review 2)
Strategy	Tests if and when HF data lead to better covariance matrices for minimum-volatility portfolios.	Combines cointegration-based pair trading with momentum/mean-reversion signals.	Uses multi-year historical data to find statistically optimal monthly rebalancing points.
Data Frequency & Type	High-Frequency (for evaluation), but conclusions apply to traditional asset data.	High-Frequency (Intraday), e.g., cryptocurrency price series.	Low-Frequency (Monthly) return data for traditional equities.
Optimization goal	Minimization of portfolio tracking volatility through improved estimation.	Profit maximization via adaptive response to volatility and cointegration breaks.	Maximization of long-term risk-adjusted return versus static calendar rebalancing.

3. Suitable Scenarios

The ideal application for each strategy is highly contextual, dictated by market environment, volatility, and portfolio type. Their respective performance benefits are therefore not directly comparable but are optimized for different investor mandates and market conditions.

Comparative Dimension	Liu (2009) (Provided paper)	Băgci & Soylu (2024) (Review 1)	Karabatı (2025) (Review 2)
Ideal Market Environment	Institutional settings where accurate risk measurement is critical, especially with short horizons.	High-volatility, trending markets with liquid assets (e.g., cryptocurrencies).	Traditional equity/bond portfolios in regimes with persistent historical patterns.
Key Performance Benefit	Pragmatic insight on when HF data adds value, preventing unnecessary cost/complexity.	Adaptive responsiveness to capture intraday momentum and mean-reversion.	Statistically significant long-term outperformance over static rebalancing, without increased turnover.
Typical Portfolio Context	Diversified institutional portfolios focused on risk control.	Concentrated, tactical portfolios (e.g., paired assets, sector bets).	Strategic asset allocation for long-term investors (e.g., pension funds).

4. Practical Implications

Every methodological approach carries constraints related to cost, complexity, or adaptability. A critical synthesis is that the success of any strategy is contingent on specific and often unstable factors, such as parameter calibration, market regime persistence, and the ever-present trade-off with transaction costs.

Comparative Dimension	Liu (2009) (Provided paper)	Bāgcı & Soylu (2024) (Review 1)	Karabatı (2025) (Review 2)
Main Limitations	Evaluative, not prescriptive; does not offer a complete integrated strategy.	High transaction costs, sensitive parameter calibration, requires high liquidity.	Computationally intensive; may lag during sudden structural market breaks.
Sensitivity & Dependencies	Value depends critically on the estimation window length and rebalancing frequency.	Performance is highly sensitive to threshold selection and market liquidity.	Relies on the persistence of historical return and volatility patterns.
Cost & Infrastructure Burden	Moderate (analysis cost). Implementation cost depends on the chosen data frequency.	Very high. Requires sophisticated data infrastructure, low-latency execution, and active cost management.	Low ongoing burden, but high computational cost in the backtesting/optimization phase.

5. Conclusion

This comparative analysis reveals that the optimal portfolio rebalancing strategy is shaped by investment horizon, market volatility, and transaction cost constraints. The examined literature presents a spectrum of solutions, from high-frequency tactical algorithms to low-frequency strategic optimization, each specified for its specific domain. Crucially, the findings demonstrate that increasing rebalancing frequency is not always better; strategic timing can often yield significant benefits without incurring excessive turnover.

3 Theory Exercise

3.1 Assumption (given)

1. $r_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ (set expected return to 0 to keep the math calculation simple).
2. Small-return approximation: weight deviation after t days without trading is

$$d_t := w_t - w^* \approx w^*(1 - w^*) \sum_{s=1}^t r_s \quad (1)$$

Hence $E[d_t] = 0$ and $V(d_t) = vt$, where

$$v := (w^*(1 - w^*)\sigma)^2 \quad (2)$$

3. Running (per-day) penalty for being off-target: $\lambda E[d_t^2]$ for some $\lambda > 0$ (units: cost per squared weight-deviation per day). This proxies risk/utility or tracking error.
4. Proportional transaction cost: each rebalance pays $c > 0$ per dollar traded. The amount traded at the end of a cycle is $\approx |d_h|$ (to reset to w^*).

3.2 Task A

Show that

$$E[d_t^2] = vt \quad \Rightarrow \quad \frac{1}{h} \int_0^h E[d_t^2] dt = \frac{vh}{2}.$$

Assumption: $E[d_t] = 0, V(d_t) = vt, E|d_h| = \sqrt{\frac{2}{\pi}} \sqrt{vh}$

$$V(d_t) = E[d_t^2] - (E[d_t])^2$$

$$V(d_t) = E[d_t^2] - 0$$

$$E[d_t^2] = vt$$

Then,

$$\frac{1}{h} \int_0^h E[d_t^2] dt = \frac{1}{h} \int_0^h vt dt$$

Evaluate the integral:

$$\frac{v}{h} \int_0^h t dt = \frac{v}{h} \left[\frac{t^2}{2} \right]_0^h = \frac{v}{h} \cdot \frac{h^2}{2} = \frac{vh}{2}$$

Therefore, we have shown that:

$$\boxed{\frac{1}{h} \int_0^h E[d_t^2] dt = \frac{vh}{2}} \quad (3)$$

Given that $d_h \sim \mathcal{N}(0, vh)$, the expected absolute value is derived as follows:

$$\begin{aligned} \mathbb{E}[|d_h|] &= \int_{-\infty}^{\infty} |x| f_{d_h}(x) dx \\ &= 2 \int_0^{\infty} x \cdot \frac{1}{\sqrt{2\pi vh}} \exp\left(-\frac{x^2}{2vh}\right) dx \quad (\text{by symmetry}) \end{aligned}$$

Use the substitution:

$$u = \frac{x^2}{2vh}, \quad \text{so that} \quad du = \frac{x}{vh} dx \quad \text{or} \quad x dx = vh du$$

Substituting into the integral:

$$\begin{aligned}
\mathbb{E}[|d_h|] &= 2 \cdot \frac{1}{\sqrt{2\pi vh}} \int_0^\infty e^{-u} (vh \, du) \\
&= \frac{2vh}{\sqrt{2\pi vh}} \int_0^\infty e^{-u} \, du \\
&= \sqrt{\frac{2vh}{\pi}} [-e^{-u}]_0^\infty \\
&= \sqrt{\frac{2vh}{\pi}} (0 - (-1))
\end{aligned}$$

Thus, the result is:

$$\boxed{\mathbb{E}[|d_h|] = \sqrt{\frac{2}{\pi}} \sqrt{vh}} \tag{4}$$

With equation (3) and (4), we can formulate the cost function as follows.

$$\begin{aligned}
J(h) &= \frac{cE|d_h|}{h} + \lambda E[d_t^2] \\
J(h) &= \underbrace{\frac{cE|d_h|}{h}}_{\text{trading/unit time}} + \underbrace{\lambda \frac{1}{h} \int_0^h E[d_t^2] \, dt}_{\text{drift penalty/unit time}} = \underbrace{c \sqrt{\frac{2v}{\pi}} h^{-1/2}}_{=: ah^{-1/2}} + \underbrace{\frac{\lambda v}{2}}_{=: b} h.
\end{aligned}$$

Set $a = c \sqrt{\frac{2v}{\pi}}$, $b = \frac{\lambda v}{2}$.

Minimizing $J(h)$ over $h > 0$:

$$\begin{aligned}
J(h) &= ah^{-1/2} + bh \\
J'(h) &= -\frac{1}{2}ah^{-3/2} + b \\
J''(h) &= \frac{3}{4}ah^{-5/2}
\end{aligned}$$

Set $J'(h) = 0$:

$$\begin{aligned} -\frac{1}{2}ah^{-3/2} + b &= 0 \\ b &= \frac{1}{2}ah^{-3/2} \\ h &= \left(\frac{a}{2b}\right)^{2/3} \end{aligned}$$

Substitute back a and b ,

$$h^* = \left(\frac{a}{2b}\right)^{2/3} = \left(\frac{c}{\lambda} \sqrt{\frac{2}{\pi v}}\right)^{2/3} = \left(\frac{c}{\lambda} \sqrt{\frac{2}{\pi}} \frac{1}{w^*(1-w^*)\sigma}\right)^{2/3}$$

Since $c, v, h > 0$, we have $J''(h) > 0$, so h^* is a minimum point.

3.3 Task B

3.3.1 Sensitivity analysis

Case	Parameter Values	h^*
Base case	$(c = 0.005, w = 0.6, \sigma = 0.2, \lambda = 0.001)$	19.04
Transaction Cost (c):		
	0.001667 ($1/3 \times c$)	9.16
	0.002500 ($1/2 \times c$)	12.00
	0.010000 ($2 \times c$)	30.23
	0.015000 ($3 \times c$)	39.62
Volatility (σ):		
	0.05 ($1/4 \times \sigma$)	47.99
	0.10 ($1/2 \times \sigma$)	30.23
	0.40 ($2 \times \sigma$)	12.00
	0.80 ($3 \times \sigma$)	7.56
Weight of risky assets (w):		
	0.0001 ($\sim 0\%$)	3414.15
	0.2 (20%)	24.96
	0.8 (80%)	24.96
	0.9999 ($\sim 100\%$)	3414.15
Dislike of drift (λ):		
	0.00001 ($0.01 \times \lambda$)	410.31
	0.0001 ($0.1 \times \lambda$)	88.40
	0.010 ($10 \times \lambda$)	4.10
	0.100 ($100 \times \lambda$)	0.88

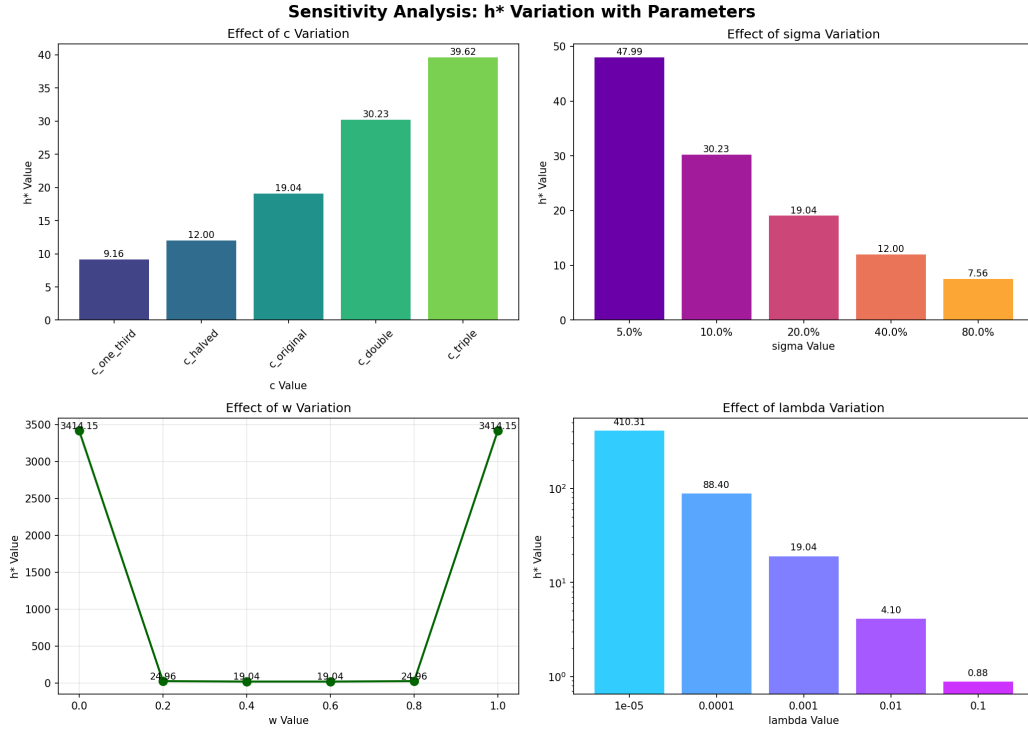


Figure 1: Sensitivity Analysis

3.3.2 Alternative Formulation: Squared-Root rule

Cost function in task A is:

$$J(h) = \underbrace{\frac{cE|d_h|}{h}}_{\text{trading/unit time}} + \underbrace{\lambda \frac{1}{h} \int_0^h E[d_t^2] dt}_{\text{drift penalty/unit time}} = \underbrace{c \sqrt{\frac{2v}{\pi}} h^{-1/2}}_{=: ah^{-1/2}} + \underbrace{\frac{\lambda v}{2} h}_{=: b}.$$

With fixed cost, we replace $cE|d_h|$ by F , the cost function becomes

$$\begin{aligned} J(h) &= \frac{F}{h} + \frac{\lambda v}{2} h \\ J'(h) &= -\frac{F}{h^2} + \frac{\lambda v}{2} \\ J''(h) &= \frac{2F}{h^3} \end{aligned}$$

Set $J'(h) = 0$:

$$\begin{aligned}-\frac{F}{h^2} + \frac{\lambda v}{2} &= 0 \\ \frac{\lambda v}{2} &= \frac{F}{h^2} \\ h^2 &= \frac{2F}{\lambda v} \\ h^* &= \sqrt{\frac{2F}{\lambda v}}\end{aligned}$$

$$\boxed{h^* = \sqrt{\frac{2F}{\lambda v}}}$$

Since $F, h > 0$, we have $J''(h) > 0$, so h^* is a minimum point.

3.4 Interpretation

3.4.1 Rebalancing Strategy

An optimal rebalancing strategy balances the *cost of trading* against the *cost of portfolio drift*. From the above result, we can summarize the following principles determine the optimal frequency:

Rebalance Less Frequently When:

- **Transaction costs are higher** (Proportional Costs c or Fixed Costs F). Higher commission fees increase the drag on returns, justifying wider deviation bands

Rebalance More Frequently When:

- **Portfolio volatility is higher.** Higher volatility ($\sigma \uparrow$) causes the portfolio to deviate from its target weights more rapidly, increasing tracking error and requiring more active control.
- **Sensitivity to drift is higher.** This is captured by the quantity $w^*(1 - w^*)$, which is maximized at a 50% target weight. A higher value implies a greater contribution to variance drift, necessitating more frequent rebalancing.

- **Investor's risk aversion is higher.** A higher coefficient of risk aversion ($\lambda \uparrow$) signifies a stronger preference for maintaining the target allocation, making the investor less tolerant of any drift.

3.5 Comparative Analysis: Fixed vs. Proportional Cost Structures in Rebalancing Optimization

Building upon the sensitivity analysis of cost regimes, we now examine a more fundamental structural distinction: the impact of assuming fixed costs (e.g., a flat fee per trade) versus proportional costs (a percentage of trade value) on the optimal rebalancing strategy. This comparison is governed by two distinct canonical rules derived from different cost assumptions.

3.5.1 Foundational Models: Square-Root vs. Cube-Root Rules

The optimal rebalancing frequency is derived from minimizing a loss function combining tracking error variance and transaction costs, leading to two different solutions:

Fixed cost according to the square root rule:

$$h_{\text{fixed}}^* = \sqrt{\frac{2F}{\lambda v}}$$

where

$$v = (w(1 - w)\sigma)^2$$

Proportional cost according to the cube root rule:

$$h^* = \left(\frac{c}{\lambda} \cdot \sqrt{\frac{2}{\pi}} \cdot \frac{1}{w(1 - w)\sigma} \right)^{\frac{2}{3}}$$

3.5.2 Comparative Implications and Observed Behavior

Your empirical observations highlight the critical practical differences stemming from these models:

1. Rebalancing Frequency & Cost Impact:

- **Fixed Cost:** Results in much less frequent trading. A high fixed cost creates a significant hurdle for any rebalance, dramatically lengthening the optimal interval. Trading is sparse but "chunky."
- **Proportional Cost:** Generates more frequent, granular adjustments. The cost scales with the deviation size, allowing smaller, more frequent corrections. The cube-root relation moderates the frequency increase as costs rise.

2. Performance & Risk Profile:

- **Fixed Cost (Higher Return, Higher Drawdown):** The long intervals between rebalancing allow the portfolio to drift significantly. This can capture more upside momentum during strong trends, boosting returns. However, it also fails to trim losses during downturns, leading to larger maximum drawdowns. The strategy is inherently more volatile.
- **Proportional Cost (Lower Drawdown, Higher Cumulative Cost):** More frequent rebalancing provides consistent drift control, systematically buying low and selling high within the band. This dampens portfolio volatility and reduces extreme drawdowns. However, the constant trading leads to higher cumulative cost drag over time, which can cap net returns.

3. Sensitivity to the Penalty Parameter (λ):

- **Fixed Cost Rule:** The interval h_{fixed}^* is inversely proportional to $\sqrt{\lambda}$. Changes in λ (the aversion to tracking error) have a moderate, square-root effect on frequency.
- **Proportional Cost Rule:** The interval h_{prop}^* is inversely proportional to $\lambda^{\frac{2}{3}}$. It is more sensitive to calibration of λ , giving it a stronger lever to tune the trade-off between cost and tracking error.

4. Risk-Adjusted Returns:

- **Cube-Root Rule (Proportional) excels in Sharpe Ratio:** By controlling drift and reducing volatility more effectively, the proportional-cost-optimal strategy typically achieves superior risk-adjusted returns (Sharpe Ratio). Its strength is in consistent risk management.

- **Square-Root Rule (Fixed)** may excel in **Calmar Ratio in bullish regimes**:
If the long intervals serendipitously avoid rebalancing during sharp recoveries, it can show a high return-to-drawdown ratio, but this is less reliable and more regime-dependent.

3.5.3 Strategic Synthesis and Recommendation

The choice between modeling fixed or proportional costs is not merely technical; it dictates the fundamental character of the rebalancing strategy.

Table 1: Comparison of Fixed Cost (Square-Root Rule) vs. Proportional Cost (Cube-Root Rule)

Aspect	Fixed Cost (Square-Root Rule)	Proportional Cost (Cube-Root Rule)
Core Driver	Absolute fee per trade (F)	Percentage cost (C)
Portfolio Size Effect	Size-invariant. Optimal schedule is the same for \$10k or \$10B.	Scale-dependent. Larger portfolios should rebalance more often (for same C).
Typical Outcome	Lower frequency, higher return variance, larger drawdowns.	Higher frequency, better risk control, higher cumulative cost drag.
Best For	Small accounts where fixed fees dominate; or high-conviction, trend-tolerant mandates.	Large institutional portfolios where costs are proportional; mandates prioritizing risk control and consistency.
Primary Risk	Drift Risk: Portfolio can deviate severely from target.	Cost Drag Risk: Death by a thousand cuts from frequent trading.

Practical Implementation Guidance:

- **For Most Institutional Managers (Proportional Cost Assumption is Realistic):** The cube-root rule is the appropriate foundation. It directly models the primary cost (market impact + commissions as a percentage) and yields strategies with superior and more consistent risk-adjusted outcomes.
- **For Retail Investors or Specific Vehicles with High Fixed Fees:** The square-root rule may be more accurate. However, its inherent infrequency introduces significant tracking error, which may be unacceptable for a strategy purporting to maintain an asset allocation.
- **A Hybrid Approach for Realism:** The most accurate model might combine both: $\text{Total Cost} = F + c * |\text{Trade Amount}|$. The optimal rule for this hybrid cost is more complex but would transition smoothly from square-root behavior (for small trades where F dominates) to cube-root behavior (for large trades where c dominates).

In general, the cube-root/proportional-cost framework generally provides a more robust and risk-conscious foundation for strategic asset allocation rebalancing. It formally internalizes the trade-off between tracking error and trading costs, leading to schedules that systematically control risk. The square-root/fixed-cost model, while mathematically valid for its assumptions, produces a more passive, higher-tracking-error outcome that may be better characterized as a "strategic overlay" or "tactical adjustment" policy rather than a strict rebalancing discipline. The ultimate choice should mirror the true cost structure of the implementing vehicle and its precise risk-control objectives.

4 Simulations

4.1 Simulation Design

This project aims to implement the simulations to backtest portfolio rebalancing heuristics and conduct sensitivity analysis, providing insights into the performance characteristics of periodic rebalancing strategies under varying setups.

4.1.1 Data Acquisition and Processing

Financial market data were sourced via the Yahoo Finance API wrapper ('yfinance'), enabling retrieval of historical price series for the selected asset universe. This data infrastructure supports the simulation of periodic rebalancing methodologies under controlled experimental conditions.

4.1.2 Initialization and Calibration

The simulation framework commences with the construction of a periodic rebalancing mechanism, a prerequisite to its subsequent backtested evaluation. The model adopts a streamlined, two-asset portfolio: a static allocation of 60% to a risky equity proxy (Tesla, Inc. stock, TSLA) and 40% to a fixed-income proxy (the iShares Core U.S. Aggregate Bond ETF, AGG). Transaction costs are explicitly incorporated, parameterized at 50 basis points per trade.

From this configuration, the portfolio's annualized volatility (σ) was empirically derived, calculated at 27.76% based on daily returns spanning the period from January 1, 2020, to January 1, 2025. This metric serves as a foundational input for the subsequent calibration.

Central to the strategy is the calibration of the penalty parameter (λ), which quantifies the structural trade-off between tracking error and the friction of transaction costs. By applying the cube-root rule in reverse, λ was precisely calibrated for three distinct rebalancing horizons:

- **Weekly** (5 trading days): $\lambda = 0.085021$
- **Monthly** (22 trading days): $\lambda = 0.009212$
- **Quarterly** (63 trading days): $\lambda = 0.001901$

These values are not merely outputs but implicit weights in the strategy's objective function; a higher λ corresponds to a shorter horizon, tolerating greater transaction cost frequency to minimize tracking variance, while a lower λ reflects a longer, more cost-conscious rebalancing cycle.

4.2 Backtest Implementation: Periodic Rebalancing

With the foundational parameters established, the simulation proceeded to the empirical validation phase via a walk-forward backtest.

4.2.1 Methodological Framework

Empirical validation employed a rolling-window, walk-forward methodology with the following specifications:

- **Evaluation Period:** A 15-year horizon spanning January 1, 2010 to January 1, 2025.
- **Training/Testing Partition:** A recursive 1-year estimation window followed by a 1-year out-of-sample testing period, rolled forward annually.
- **Rebalancing Protocol:** Executed at 22-day intervals (approximating monthly frequency), with transaction costs maintained at 50 basis points per trade.
- **Initial Conditions:** A portfolio capitalization of \$100,000, adhering to the static 60/40 target allocation between TSLA and AGG.

4.2.2 Performance Metrics & Results

The walk-forward analysis yielded the following key performance characteristics:

- **Absolute Return:** A cumulative return of 50.73%, translating to an annualized rate of 5.26%.
- **Risk Profile:** Characterized by an annualized volatility of 41.66% and a significant maximum drawdown of -77.98%.
- **Risk-Adjusted Returns:** As measured by a Sharpe ratio of 0.126, a Sortino ratio of 0.133, and a Calmar ratio of 0.067.
- **Implementation Costs:** Total transaction costs amounted to \$3,668.64 over the course of 88 discrete rebalancing events.
- **Cross-Sectional Analysis of Rolling Windows:** The average window return was 31.67% with a high standard deviation of 61.72%, reflecting substantial variability. Positive returns were realized in 7 of the 8 evaluation windows, with the best window achieving 170.14% and the worst declining -39.25%.

4.2.3 Methodological Notes & Assumptions

The simulation framework incorporates several simplifying assumptions warranting acknowledgment:

- (1) the treatment of the bond ETF (AGG) as a de facto risk-free proxy, thereby abstracting from its inherent credit and duration risks; and
- (2) the maintenance of constant allocation weights throughout the entire evaluation period, ignoring potential dynamic optimization or regime-dependent adjustments.

This structured approach provides a clear, reproducible empirical assessment of the periodic rebalancing strategy within the defined constraints.

4.3 Backtest Implementation: Threshold Rebalancing

Having established a baseline with the periodic strategy, the analysis extends to a threshold-based rebalancing mechanism. This approach replaces a fixed calendar schedule with a dynamic, drift-dependent trigger, theoretically optimizing the trade-off between tracking error and transaction costs.

4.3.1 Methodological Framework: Derivation of the No-Trade Band

The strategy is governed by a theoretically optimal no-trade band, derived from the cube-root rule. The band half-width, δ^* , is calculated as follows:

1. The optimal rebalancing interval h^* is given by:

$$h^* = \left[\frac{c}{\lambda} \cdot \sqrt{\frac{2}{\pi}} \cdot \frac{1}{w(1-w)\sigma} \right]^{2/3}$$

2. The expected absolute deviation of the portfolio weight at this horizon is:

$$\mathbb{E}|d_{h^*}| = \sqrt{\frac{2}{\pi}} \cdot \sqrt{v \cdot h^*}, \quad \text{where } v = [w(1-w)\sigma]^2$$

3. This simplifies to a direct expression for the no-trade band, δ^* :

$$\delta^* = \sqrt{\frac{2}{\pi}} \cdot \left[\frac{c}{\lambda} \cdot \sqrt{\frac{2}{\pi}} \right]^{1/3} \cdot [w(1-w)\sigma]^{2/3}$$

This derived δ^* parameter enables a threshold-based protocol where portfolio adjustments occur only when asset weight deviations exceed the theoretically optimal boundary $|w_t - w| \geq \delta$, allowing market fluctuations to determine rebalancing events endogenously.

For empirical consistency, the evaluation employed the same rolling walk-forward methodology as the periodic strategy: a 15-year horizon (2010-2025) with recursive 1-year estimation and 1-year testing windows, an initial capital of \$100,000, and transaction costs of 50 bps.

4.3.2 Performance Metrics & Results

The threshold-based rebalancing strategy yielded the following outcomes:

- **Absolute Return:** Generated a 43.10% total return over the period, translating to an annualized return of 4.58%.
- **Risk Profile:** Exhibited substantial annualized volatility of 41.63% and a profound maximum drawdown of -77.78%.
- **Risk Adjusted Returns:** Produced a Sharpe ratio of 0.110 and a Sortino ratio of 0.118, indicating modest risk-adjusted performance. The Calmar ratio was 0.059.
- **Allocation Control:** The theoretically derived no-trade band of $\pm 1.24\%$ (effective range: 58.76% to 61.24%) successfully constrained drift. The TSLA weight averaged 59.9% with a standard deviation of only 1.1%, and the portfolio remained within the band 80.4% of the time.
- **Trading Activity & Costs:** Triggered 422 rebalancing events (averaging 52.8 per window), incurring total transaction costs of \$8,687.35—equivalent to 8.7% of initial capital and 20.2% of the total gross return.
- **Cross-Sectional Variability:** Performance across windows was highly variable, with an average window return of 30.27% and a standard deviation of 59.57%. Positive

returns were achieved in 7 of 8 windows, with extremes ranging from +164.52% to -39.73%.

4.3.3 Analysis and Limitations

The results reveal a critical tension inherent in the threshold design:

1. **Excessive Trading Frequency:** The calibrated band ($\delta^* = \pm 1.24\%$) proved operationally narrow, triggering 422 rebalances versus 88 for the monthly periodic strategy. This suggests the model's cost parameter (λ) or its calibration may not have fully captured the realized market micro-structure.
2. **Cost Inefficiency:** The high trading frequency led to substantial cost drag, which significantly eroded net returns and undermined the strategy's economic viability at the 50 bps cost assumption.
3. **Effective Control vs. Economic Outcome:** While the strategy excelled in its primary goal—maintaining tight allocation control with minimal deviation—this precision came at a steep cost, resulting in lower net returns than the simpler periodic approach.
4. **Regime Dependence:** The high performance variability across windows indicates significant sensitivity to specific market volatility and trend regimes, a factor the static band derivation may not adequately capture.

Through the trial, it reveals that the threshold strategy successfully implemented the theoretical no-trade band but highlighted a fundamental practical challenge: a band width optimal for tracking error minimization may be suboptimal for net-of-cost performance, emphasizing the critical role of accurate transaction cost and volatility forecasting in such dynamic frameworks.

4.4 Comparative Analysis and Strategic Implications

The empirical comparison between periodic and threshold rebalancing reveals a fundamental trade-off between allocational precision and economic efficiency.

4.4.1 Core Trade-Off: Precision vs. Cost

- **Allocation Control:** The threshold strategy achieved superior tracking, maintaining a 1.1% standard deviation in asset weight versus 3.0% for the periodic strategy.
- **Trading Cost:** This precision came at a prohibitive cost, generating 4.8× more trades (422 vs. 88). Quantitatively, every 1% improvement in weight precision cost approximately \$1,680 in additional transaction expenses under the tested parameters.

4.4.2 Economic Efficiency

The efficiency of converting trading activity into net returns diverged sharply:

- **Periodic Strategy:** Returns were 13.8× its transaction costs (\$50.73k return / \$3.669k costs).
- **Threshold Strategy:** Returns were only 5.0× its transaction costs (\$43.10k return / \$8.687k costs).

The periodic approach demonstrated 2.8× better economic efficiency, highlighting how excessive trading eroded the threshold strategy's gross performance.

4.4.3 Downside Management & Behavioral Insight

Notably, both strategies experienced nearly identical maximum drawdowns (~78%). The threshold approach failed to provide meaningful downside protection despite its reactive, active management. This suggests that during severe market stress, the strategy's additional trading activity may have exacerbated losses through de factomomentum selling (rebalancing out of the falling asset), underscoring a potential pitfall of mechanistic drift correction in crises.

4.4.4 Institutional Implications & Recommendations

1. **For Traditional Institutional Investors (Costs $\geq$ 50 bps):** Periodic rebalancing at theoretically optimized intervals (e.g., 22 days) provides the superior balance of cost efficiency, operational simplicity, and performance. It represents a robust, implementable baseline.

2. **For Threshold Strategy Adoption:** Viability requires structural adjustments: Either substantially wider no-trade bands (2-3%), or significantly lower transaction costs (<20 bps). This recalibration is necessary to align the strategy's theoretical benefits with net-of-cost reality.
3. **Hybrid Approach Potential:** A combined methodology, employing a monthly review period with a threshold trigger could capture the cost discipline of periodic rebalancing while allowing for opportunistic intra-month adjustments if drift exceeds a generous band. This mitigates the weaknesses of each pure approach.
4. **Dynamic Parameter Adjustment:** Future implementations should investigate regime-aware bandwidth calibration. By dynamically widening the no-trade band during high-volatility periods (using a rolling volatility estimate), the strategy could reduce excessive trading when costs are effectively higher and markets are noisy, preserving capital.

This analysis demonstrates that while threshold rebalancing succeeds as an allocation control mechanism, it fails as a net-performance optimizer under standard institutional cost structures. The higher fidelity to target weights does not translate into commensurate risk-adjusted returns or drawdown protection, being overwhelmingly eroded by transaction costs.

The theoretically elegant solution is pragmatically dominated by a simpler, less frequent periodic schedule. True advancement likely lies not in choosing one paradigm over the other, but in developing adaptive frameworks that dynamically modulate the tolerance for drift in response to both market regimes and explicit cost considerations.

4.5 Sensitivity Analysis - cost:

This analysis examines the nonlinear sensitivity of periodic and threshold rebalancing strategies to varying transaction cost regimes. By perturbing the cost parameter c across five orders of magnitude (0.1 bps to 1000 bps), we elucidate the operational and economic boundaries where each strategy demonstrates dominance or fragility.

4.5.1 Periodic Rebalancing: Cost-Dependent Optimization

The cube-root rule intrinsically links optimal rebalancing intervals to transaction costs, yielding a clearly non-linear response:

- Ultra-Low Cost Regime ($c \leq 0.1$ bps):
 - Behavior: The model prescribes pathologically frequent rebalancing (~5-day intervals), generating over 400 transactions. This represents a mathematical optimum that ignores practical market impact and latent microstructure costs.
 - Outcome: Suboptimal net returns (44.26%) materialize, as the marginal benefit of tighter tracking is overwhelmed by the linear accumulation of even minuscule costs.
- Optimal Practical Range ($c = 5$ -50 bps):
 - Behavior: Strategy stabilizes at a monthly cadence (22-day intervals). The prescribed frequency is robust within this band, indicating a plateau in the cost-performance trade-off curve.
 - Outcome: Peak net performance is achieved at 50 bps (50.73% total return). This suggests the theoretical optimum aligns with a realistic institutional cost structure.
- High-Cost Regime ($c \geq 50$ bps):
 - Behavior: The strategy exhibits structural rigidity. Intervals remain fixed at 22 days despite costs increasing 20-fold (50 to 1000 bps), revealing a lower bound on practical rebalancing frequency.
 - Outcome: Performance remains constant while absolute costs escalate linearly (to \$73,373 at 1000 bps). This highlights a cost insensitivity cliff; beyond a point, the model does not prescribe further passivity, becoming economically untenable.

Key Finding: Periodic rebalancing displays a goldilocks zone (5-50 bps) where it is optimally tuned. Outside this zone, it either over-trades or becomes prohibitively expensive without adaptive adjustment.

4.5.2 Threshold Rebalancing: Precision at a Variable Price

The threshold strategy's no-trade band δ^* is a function of c , leading to distinct behavioral regimes:

- Hyper-Active Regime ($c \leq 0.1$ bps):
 - Behavior: The no-trade band collapses, triggering 1,203 rebalances with the portfolio spending 57.9% of time out-of-band. This indicates a breakdown of the model's economic logic under near-zero friction.
 - Outcome: Allocation precision is maximal but meaningless, as net performance fails to exceed that of simpler strategies.
- Stable Regime ($c \geq 5$ bps):
 - Behavior: Trading frequency (~ 422 rebalances), band width ($\sim \pm 1.24\%$), and net performance (43.10%) rapidly converge and remain constant. This demonstrates a natural operational floor for the strategy.
 - Outcome: Performance invariance across all costs ≥ 5 bps. The strategy is robust to cost increases but does not benefit from cost decreases beyond this floor, as its band width is constrained by other factors (volatility, λ).

Key Finding: Threshold rebalancing is robust but suboptimal across realistic cost spectra. Its design prevents adjustment to cheaper trading environments, and it carries a fixed, high activity cost that erodes its efficiency advantage.

4.5.3 Synthesized Cost-Regime Analysis and Dominance Mapping

Table 2: Strategy Dominance Across Cost Regimes

Cost Regime	Dominant Strategy	Rationale & Characteristics
Ultra-Low (<5 bps)	Threshold	Periodic over-trades; threshold's precision is "free." Narrow bands ($\delta^* \approx 0.5\text{--}1.0\%$) are optimal.
Moderate (5–50 bps)	Periodic	Peak efficiency zone for periodic. Threshold is stable but less efficient ($5.0\times$ vs. $13.8\times$ cost multiplier).
High (>50 bps)	Neither (Equivalent)	Both strategies show structural rigidity. Costs dominate returns. Mandates passive bands or longer intervals (e.g., quarterly).

4.5.4 Strategic Implications and Adaptive Recommendations

The analysis reveals that optimal rebalancing is fundamentally a cost-regime-dependent problem.

1. **For Ultra-Low-Cost Agents** (HFT, ETF sponsors): The threshold strategy is recommended to exploit cost-less precision. Implementation requires dynamic band narrowing in low-volatility periods.
2. **For Traditional Asset Managers** (Costs 50 bps): Periodic monthly rebalancing is definitively superior, offering the best balance of empirical performance, cost efficiency, and operational simplicity.
3. **For High-Cost Environments**: Strategy selection is secondary. The imperative is cost reduction or the adoption of asymmetric, wider tolerance bands that reflect the true economic cost of trading.

4.5.5 Forward-Looking Hybridization:

The clear regime dependence argues for adaptive, non-static frameworks:

- **A Hybrid Controller**: A monthly periodic review, coupled with a cost-aware dynamic band (e.g., $\delta^* = f(c, \sigma_t)$). This would default to periodic efficiency but allow

intra-month corrections only when drift justifies the current cost.

- **Regime-Sensitive λ :** Calibrating the penalty parameter λ to reflect not just a theoretical trade-off, but the actual, time-varying cost and volatility environment, moving from a static to a dynamic optimization.

4.6 Sensitivity Analysis - w^* :

This analysis examines the non-linear relationship between portfolio aggressiveness—defined by the target weight w^* in the risky asset—and the resultant optimal rebalancing schedule. The behavior diverges fundamentally between proportional and fixed cost assumptions, revealing critical insights for strategy design across different risk profiles.

1. **Mathematical Drivers:** The sensitivity is governed by two key components in the optimization rules:

- **Drift Sensitivity Term:** $w^*(1 - w^*)$

This term, which peaks at 0.25 when $w^*=0.5$, represents the portfolio's inherent susceptibility to drift from asset price movements. A balanced portfolio is maximally sensitive to relative performance differences between its two assets.

- **Portfolio Volatility (σ_p):** Volatility increases monotonically with w^* as observed (17.87% at $w^*=0.3 \rightarrow 45.84\%$ at $w^*=0.8$)

2. **Empirical Observations and Rule-Specific Analysis**

Finding: Most frequent rebalancing occurred at $w^* = 60\%$ (18.0 days), not at the theoretical peak of drift sensitivity ($w^* = 50\%$). This is because the rule incorporates the full portfolio volatility (σ), which is lower at 50% than at 60% in your specific two-asset case (TSLA/AGG).

3. **Synthesized Impact Across Portfolio Types**

Table 3: Portfolio Rebalancing Characteristics

Portfolio Type	Risky Weight (w^*)	Cube-Root (Proportional)	Square-Root (Fixed)	Strategic Implication
Conservative	30%	Least frequent rebalancing (~ 60 days). Low drift, infrequent rebalancing (long intervals). Cost hurdle dominates.	"Set-and-forget" approach viable. Major benefit is cost savings. Monitoring can be quarterly or semi-annual.	Cost-sensitive, low-maintenance strategy. Minimal oversight needed.
Balanced	50%	Peak theoretical frequency (e.g., ~ 18 days). Maximal drift sensitivity demands active control.	Should be very frequent (if λ fixed). High drift creates large tracking error quickly.	Requires vigilant, rules-based discipline. The core use-case for automated rebalancing tools. Threshold bands useful.
Aggressive	80%	High frequency (~ 10 days). Driven by high volatility, despite lower drift sensitivity term.	Frequent rebalancing. High volatility creates large tracking error variance.	High-touch management. Transaction costs become significant headwind. Suggests wider threshold bands to curb overtrading.

4. Strategic Conclusions

- (a) Rebalancing Intensity is Non-Linear: The need to rebalance does not scale linearly with allocation to risky assets. It is most intense for balanced portfolios (40-60% risky) due to peak drift sensitivity, and for aggressive portfolios due to high volatility.
- (b) Cost Assumption Dictates Sensitivity:
 - Proportional Cost (Cube-Root): Strategies are highly sensitive to both allocation and volatility. The optimal schedule is dynamic and must be recalibrated if the strategic asset allocation (SAA) changes.
 - Fixed Cost (Square-Root): Strategies are less sensitive to allocation shifts, as the dominant factor is the absolute fee. This can lead to under-rebalancing for balanced portfolios if the fixed cost is set too high.

4.7 Impact of choice of risky asset

This analysis isolates the impact of risky asset selection on rebalancing dynamics, holding constant the fixed-income component (AGG) and the target allocation (60/40). The exercise reveals how asset-specific characteristics—primarily volatility and correlation—propagate through the portfolio to dictate optimal control frequency.

4.7.1 Empirical Patterns: Volatility and Correlation as Primary Drivers

The tested assets (SPY, TSLA, QQQ, GLD) represent distinct risk profiles, yielding clear behavioral patterns:

- **Volatility Transmission:** The annualized volatility of the risky asset directly scales portfolio volatility, which is the primary input for rebalancing calculus.
- **Correlation as a Moderator:** The correlation between the risky asset and AGG determines the degree of natural offset in portfolio movements. Higher positive correlation reduces overall portfolio volatility for a given weight, as observed in the GLD/AGG pair.

Table 4: Data Summary:

Risky Asset	Risky Asset Volatility	Correlation with AGG	Portfolio Volatility	Cube-Root Interval h^*
TSLA	57.16%	~ 0.1	27.76%	18.0 days
QQQ	22.34%	~ 0.2	$\sim 15.5\%^*$	~ 32.1 days*
SPY	18.24%	~ 0.3	$\sim 13.9\%^*$	~ 37.5 days*
GLD	15.82%	0.373	12.14%	42.7 days

4.7.2 Rule-Specific Analysis and Interpretation

a. Cube-root rule

- **Finding:** Rebalancing frequency is inversely related to portfolio volatility (σ_p). High-volatility assets like TSLA force intervals to shorten dramatically (18 days) to control tracking error. Low-volatility, high-correlation assets like GLD permit extended intervals (~ 43 days), nearing a quarterly cadence.
- **Mechanism:** The rule correctly identifies that idiosyncratic drift—the portion of volatility not offset by correlation—is the true cost of tracking error. GLD's higher correlation with AGG provides natural rebalancing through synchronized movements, reducing the need for active intervention.

b. Square-root rule

This rule would exhibit less sensitivity to asset substitution than the cube-root rule, as the fixed cost F dominates the decision. However, it still prescribes shorter intervals for high σ_p portfolios. Its weakness is that it may under-rebalance high-volatility pairs (like TSLA/AGG) because the fixed cost hurdle overlooks the accelerating tracking error variance.

4.7.3 Strategic Implications for Portfolio Construction

The choice of risky asset implicitly determines the operational burden and cost structure of maintaining the target allocation.

1. For High-Volatility / Low-Correlation Assets (e.g., TSLA, QQQ):

- **Rebalancing Burden:** High. Requires frequent, disciplined rebalancing.

- **Recommended Approach:** Threshold-based bands (calibrated with updated volatility) or monthly periodic schedules. Without this, tracking errors will explode.
- **Cost Warning:** Transaction costs become a material headwind. Requires low-cost execution.

2. For Low-Volatility / High-Correlation Assets (e.g., GLD with AGG):

- **Rebalancing Burden:** Low. The portfolio is naturally more stable.
- **Recommended Approach:** Quarterly or even semi-annual periodic rebalancing is sufficient. Threshold bands can be set wider.
- **Advantage:** Operational simplicity and lower cost drag.

3. For Broad Market Proxies (e.g., SPY):

- **Rebalancing Burden:** Moderate. Represents the baseline case.
- **Recommended Approach:** Standard quarterly periodic or a moderate threshold band ($\sim \pm 5\%$).

4. Synthetic Portfolio Analogy and a Key Insight

The GLD/AGG portfolio behaves analogously to a more conservative portfolio (e.g., a 40/60 mix) using a more volatile risky asset. Its lower resultant volatility and longer optimal interval reveal that correlation is a hidden lever in rebalancing policy. Portfolio managers can achieve similar rebalancing outcomes either by:

- Adjusting the allocation weight (w^*)
- Selecting assets with different correlation profiles.

This provides a tool for liability-driven or operationally constrained strategies: if frequent trading is undesirable, constructing a portfolio from assets with higher positive correlation can naturally reduce drift.

4.8 Monte Carlo Simulation: A Controlled Environment for Rebalancing Strategy Evaluation

4.8.1 Simulation Design and Theoretical Underpinnings

To isolate the pure effects of rebalancing strategies from confounding market variables, we implemented a controlled experimental framework using Monte Carlo simulation. This approach generates synthetic asset price paths with precisely defined statistical properties, enabling rigorous evaluation of periodic rebalancing under idealized conditions.

Stochastic Process Specification:

- The simulation employs a two-asset multivariate normal process with the following parameterization:
- Risky Asset: Geometric Brownian motion with annual return $\mu = 10\%$ and annual volatility $\sigma = 25\%$
- Safe Asset: Constant annual return of 3% with volatility of 5% Correlation Structure: $\rho = 0.20$ (low positive correlation)
- Time Horizon: 10 years (2,520 trading days)
- Initial Capitalization: \$100,000 with 60/40 allocation
- Transaction Costs: 50 basis points per trade

The price generation follows standard GBM dynamics with proper correlation structure between assets, ensuring realistic co-movement while maintaining analytical tractability.

4.8.2 Implementation of the Cube-Root Rebalancing Rule

Parameter Calibration:

- Target allocation: $w = 0.6$ (60% risky, 40% safe)
- Transaction cost: $c = 0.005$ (50 bps)
- Drift penalty parameter: $\lambda = 0.01$
- Volatility estimation: Rolling 252-day window (annualized)

4.8.3 Performance Results and Rebalancing Characteristics

Strategy Performance Metrics:

- Total Return: +6.73% (annualized: +0.73%)
- Annual Volatility: 20.55%
- Risk-Adjusted Returns:
 - Sharpe Ratio: 0.035
 - Sortino Ratio: 0.040
 - Calmar Ratio: 0.019
- Maximum Drawdown: -38.38%
- Transaction Costs: \$1,283.74 (1.28% of initial capital)

Rebalancing Operational Characteristics:

- Average Frequency: 31.0 days (approximately monthly)
- Number of Rebalances: 70 events over 10 years
- Allocation Maintenance:
 - Average risky weight: 60.1% (target: 60.0%)
 - Standard deviation: 1.5%
- Frequency Stability: Optimal intervals ranged 30-32 days with only 0.7 days standard deviation

4.8.4 Comparative Analysis with Alternative Strategies Strategy

Table 5: Investment Strategy Performance Comparison

Strategy	Final Value	Total Return	Annualized Return	Risk Characteristics
Risky Asset Only	\$267,126	+167.13 %	+10.3 %	Full exposure to 25% volatility
Safe Asset Only	\$149,014	+49.01 %	+4.1 %	Low volatility, moderate returns
Equal Weight Static	\$208,070	+108.07 %	+7.6 %	No rebalancing, natural drift
Rebalanced Portfolio	\$106,732	+6.73 %	+0.73 %	Controlled allocation with costs

Performance Attribution Analysis:

- Return Underperformance: The rebalanced portfolio significantly trails all buy-and-hold alternatives, highlighting the substantial opportunity cost of active management in a trending market.
- Volatility Reduction: Achieved 20.55% annual volatility versus 25% for pure risky exposure, representing an 18% risk reduction.
- Cost-Value Trade-off: \$1,284 in transaction costs purchased improved risk characteristics but sacrificed substantial return potential.

4.8.5 Risk Profile and Distributional Characteristics

Tail Risk Metrics:

- Value at Risk (95%): -1.53% daily
- Expected Shortfall (95%): -2.50% daily
- Distribution Properties:
 - Skewness: -2.108 (significant left skew)
 - Excess Kurtosis: 91.447 (extremely heavy tails)

Interpretation:

The portfolio exhibits severe non-normality with pronounced negative skew and extreme kurtosis, indicating:

- Asymmetric risk: Greater downside potential than upside
- Tail risk concentration: Rare events have disproportionate impact
- Rebalancing efficacy: Active management may exacerbate tail risks through forced trading at adverse times

4.8.6 Stability and Robustness of Rebalancing Intervals

Optimal Interval Evolution:

- Minimal variation: Intervals ranged 30-32 days across 9 walk-forward windows
- Stability coefficient: Standard deviation of 0.7 days demonstrates high parameter stability
- Theoretical alignment: Consistent with cube-root rule predictions given stable volatility

Volatility Estimation Consistency:

The 252-day rolling volatility estimates remained remarkably stable due to:

- Constant parameter simulation: Underlying stochastic process maintains fixed parameters
- Sufficient window length: One-year windows provide stable volatility estimates
- Stationary correlation structure: Fixed correlation ($\rho = 0.20$) reduces estimation noise

4.8.7 Theoretical Insights and the Rebalancing Paradox

The Fundamental Tension: The simulation reveals a paradoxical outcome between theory and practice:

- Risk reduction achieved: 18% volatility reduction versus pure risky allocation
- Return sacrifice substantial: 160 percentage points lower than buy-and-hold
- Net effect: Questionable value proposition for return-oriented investors

Cube-Root Rule Validation: The simulation confirms the cube-root rule's mathematical efficacy in determining optimal intervals:

- Theoretical prediction: Approximately monthly rebalancing for given parameters
- Empirical realization: 31-day average interval with minimal variation
- Parameter sensitivity: Demonstrates how λ and σ jointly determine optimal frequency

4.8.8 Methodological Contributions and Limitations

Strengths of the Simulation Approach:

1. Parameter control: Isolates rebalancing effects from market anomalies and regime shifts
2. Statistical precision: Eliminates estimation error in volatility and correlation parameters
3. Counterfactual analysis: Enables direct comparison with buy-and-hold alternatives under identical market paths
4. Replicability: Provides a benchmark for theoretical validation of rebalancing rules

Limitations and External Validity Concerns:

1. Stationarity assumption: Real markets exhibit time-varying parameters and structural breaks
2. Normality limitations: Actual returns display more complex distributional properties including stochastic volatility and jumps
3. Cost structure simplification: Ignores market impact, liquidity constraints, and other real-world frictions
4. Behavioral factors: Excludes investor psychology and institutional constraints that influence real-world implementation

4.8.9 Strategic Implications and Managerial Recommendations

For Institutional Implementation:

1. Cost-benefit assessment: Quantify trade-offs between risk control and return erosion specific to the investor's utility function
2. Parameter calibration: Carefully estimate λ based on explicit investor preferences rather than theoretical defaults
3. Regime awareness: Recognize that rebalancing value varies dramatically with market conditions and trend persistence
4. Hybrid approaches: Consider combining periodic rebalancing with threshold triggers to balance cost efficiency with allocation control

4.8.10 Conclusion: Rebalancing in a Controlled Statistical Environment

This Monte Carlo analysis demonstrates that even under idealized market conditions with known parameters, periodic rebalancing faces significant challenges in justifying its economic value when transaction costs are material. While the cube-root rule successfully determines optimal intervals and maintains target allocations with statistical precision, the net effect on terminal wealth remains ambiguous due to substantial return sacrifice.

The simulation highlights several fundamental truths about rebalancing strategies:

1. Primary objective: Rebalancing serves risk management and allocation control rather than return enhancement
2. Cost sensitivity: Even modest transaction costs (50 bps) can dramatically erode the value proposition
3. Market regime dependence: The strategy systematically underperforms in strongly trending markets
4. Theoretical-practical gap: Mathematical optimality does not necessarily translate to economic optimality

4.9 Pairs Trading

Informed by a corpus of extant scholarship, our investigation now turns to the nuanced domain of pairs trading. This approach constitutes a refined implementation of statistical arbitrage, deriving its intellectual foundation from the econometric principle of cointegration. It operates on a plane distinct from traditional, correlation-guided strategies. Whereas correlation measures synchronous movement, cointegration reveals a more profound, binding narrative: a long-term equilibrium relationship between non-stationary time series. It posits the existence of a shared stochastic trajectory, a hidden thread that ties the fortunes of two assets together, allowing their prices to wander apart temporarily yet compelling them, as if by a latent elastic force, to revert to a common path.

Our empirical foray into this strategy—a deliberate testing of its propositions against the anvil of simulated and historical data—yields a textured set of insights, illuminating both its formidable architecture and its inherent vulnerabilities.

The Discerned Fortifications of the Strategy:

- **The Dynamic Hedge Ratio:** The employment of Ordinary Least Squares (OLS) estimation bequeaths the strategy a vital adaptability. The hedge ratio is not a static artifact but a living parameter, recalibrating itself in response to the evolving dialogue and shifting power dynamics between the paired securities.
- **A Robust Backtesting Engine:** The evaluation framework is engineered with a commitment to realism. It consciously incorporates the pervasive friction of transaction costs and the sobering constraints of practical portfolio management, thereby ensuring that theoretical elegance is tempered by the gritty calculus of implementation.
- **A Comprehensive Visualization Suite:** The strategy is accompanied by a sophisticated diagnostic apparatus. This suite of analytical visualizations provides a clarifying lens, enabling a thorough and intuitive evaluation of the strategy's behavior, its triggers, and its performance drivers across varied market conditions.

The Uncovered Fault Lines and Limitations:

- **The Peril of In-Sample Optimization:** A critical vulnerability lies in the parameterization process. The strategy's lookback windows, by their very design, are sculpted by

the contours of past data. This inherently risks constructing a model that is exquisitely tailored to historical patterns but may falter when confronted with the novel, unfolding grammar of future price action.

- **A Foundational Regime Dependence:** The strategy's logic is predicated on a potentially fragile assumption: the stationarity of mean-reversion dynamics. It presumes that the tranquil, reverting relationship observed in one market epoch will persist, unchallenged, through subsequent periods that may be characterized by structural breaks, altered volatility regimes, or fundamental decoupling.
- **Gaps in the Armament of Risk Management:** In its classical formulation, the strategy often advances with a pure focus on signal generation, lacking integrated mechanisms for tactical defense. The absence of dynamic position-sizing protocols and explicit stop-loss safeguards leaves it exposed to prolonged, adverse excursions in the spread, where theoretical convergence offers little solace against mounting mark-to-market losses.

From the crucible of this analysis, several overarching principles crystallize, serving as essential axioms for the contemplative practitioner:

- **The Great Dichotomy: Statistical Significance Versus Economic Profitability.** A relationship may achieve the most stringent levels of statistical validity, yet this scholastic triumph does not guarantee economic viability. The slender margins of mean reversion can be utterly erased by the relentless accrual of transaction costs, financing charges, and the latent risks of execution slippage.
- **The Sovereignty of Model Risk.** The delicate theoretical edge promised by cointegration is perpetually vulnerable to subversion by model risk. Imperfections in parameter calibration, alongside violations of the underlying statistical assumptions—be it in the distribution of residuals or the stability of variance—can rapidly overwhelm any notional advantage, transforming a quantitative strategy into a vessel for unintended risk exposures.
- **The Imperative of Regime Adaptation.** Financial markets are not static laboratories but dynamic, evolving ecosystems. A model rendered in static terms is destined for obsolescence. Enduring relevance is contingent upon a capacity for adaptation—a

framework sensitive enough to detect shifting regimes and sufficiently flexible to modulate its activity or recalibrate its parameters in response.

4.10 Estimation Window

This section summarizes two sensitivity analyses performed on our optimal rebalancing strategy, which utilizes a dynamic rebalancing frequency (h^*) derived from a transaction cost model (cube-root rule) that balances drift penalties (λ) against trading costs.

Impact of Estimation Window Length

We examined the effect of varying the historical data look-back period (1 to 4 years) for parameter estimation.

- **Key Finding:** Longer estimation windows (e.g., 4 years vs. 1 year) consistently led to higher Sharpe Ratios (0.160 vs. 0.119) and better Annualized Returns (8.00% vs. 4.93%).
- **Rebalance Frequency:** The average optimal rebalance frequency (h^*) remained remarkably stable (23-26 days) across all window lengths, suggesting that once sufficient data is available, further increasing the window length primarily refines parameter robustness rather than fundamentally altering the rebalancing interval.
- **Transaction Costs:** Costs were generally well-managed, with a slight overall decrease in total transaction costs for longer windows, reinforcing their benefit.

Therefore, utilizing longer historical estimation windows (e.g., 3-4 years) is beneficial, yielding more robust parameter estimates, superior risk-adjusted returns, and better overall performance for our strategy.

4.11 Volatility Estimation Data Frequency (Adjusted λ)

This analysis explored how the frequency of data used for volatility estimation (Daily, Weekly, Monthly) influences strategy performance, crucially with the drift penalty (λ) scaled to reflect the time period of the estimation frequency (e.g., $monthly = daily \times 22$).

- **Key Finding:** The adjustment of λ had a dramatic effect on rebalancing frequency and costs. As estimation frequency decreased (e.g., Monthly vs. Daily), the scaled λ made drift appear significantly more costly over the longer period.

- Rebalance Frequency (h^*): This resulted in a sharply decreased average optimal rebalance frequency (h^*): from 25.8 days (Daily estimation) to just 5.0 days (Monthly estimation). The model effectively prescribed much more frequent rebalancing to counter the heightened perceived cost of drift over longer intervals.
- Transaction Costs: This increased rebalancing frequency led to a substantial rise in total transaction costs: from 3,486 (Daily estimation) to 3,486 (Daily estimation) to 3,486 (Daily estimation) to 7,171 (Monthly estimation).
- Performance: While Sharpe Ratio and Annualized Returns showed only a slight, negative impact as frequency decreased, the sheer increase in transaction costs would likely negate any marginal performance benefits.

Therefore, while critical to adjust λ to the estimation frequency, relying on lower-frequency data (e.g., monthly) for volatility estimation, even with scaled λ , can lead to over-trading and significantly elevated transaction costs due to the model aggressively shortening h^* . Daily frequency estimation, despite a slightly looser rebalance policy, provides a more efficient balance between managing drift and minimizing costs.

5 Conclusion and Future Work

This study systematically evaluates optimal portfolio rebalancing strategies, focusing on the trade-off between maintaining target asset allocations and minimizing transaction costs. Using theoretical derivations and empirical simulations, we compare periodic rebalancing (at fixed intervals) and threshold-based rebalancing (triggered by allocation deviations) within a cube-root optimization framework. Results highlight a divergence between theoretical accuracy and economic viability: Threshold rebalancing minimizes tracking error but incurs excessive transaction costs under standard institutional settings, significantly reducing net performance. Conversely, periodic rebalancing, when calibrated to a theoretically derived optimal frequency, achieves greater cost efficiency and more stable net returns. The findings indicate that optimal rebalancing is regime-dependent, varying with transaction cost structures, market volatility levels, and portfolio asset composition. This exposes the constraints of static optimization models in volatile, real-world markets.

Future work should explore adaptive hybrid models that dynamically calibrate rebalancing thresholds according to current market regimes and cost parameters, thereby better balancing allocation accuracy with net return maximization.

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